Continuous Distributions

1 Random Variables of the Continuous Type

Definition 1.1 (Probability Density Function)**.**

A function $f\colon \mathbb{R} \to [0,1]$ is called the **probability density function** (pdf) of a CRV X with range S if it satisfies:

- $f(x) > 0$ for $x \in S$, and $f(x) = 0$ for $x \notin S$
- $\displaystyle\int_S f(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$
- $\mathbb{P}(X \in A) = \displaystyle\int_A f(x)\,\mathrm{d}x$, where $A \subseteq S$

Then S is called the **support** or **space** of $f(x)$.

Remark. Pdf may not be continuous or bounded.

Remark. Different from pmfs, if f is a pdf of a CRV X, then, generally,

$$f(x) \neq \mathbb{P}(X = x)$$

Definition 1.2 (Cumulative Distribution Function)**.**

The function $F\colon \mathbb{R} \to [0,1]$ defined by

$$F(x) := \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t \qquad -\infty < x < \infty$$

is called the **cumulative distribution function** (cdf) or **distribution function** of X.

Remark.

$$F(b) - F(a) = \mathbb{P}(a < X < b) = \mathbb{P}(a \leqslant X < b) = \mathbb{P}(a < X \leqslant b) = \mathbb{P}(a \leqslant X \leqslant b) = \int_a^b f(t)\,\mathrm{d}t$$

$$f(x) = F'(x) \quad \text{for those } x \text{ at which } F(x) \text{ is differentiable}$$

Theorem 1.1.

For two CRVs X and Y, if $Y = g(X)$, where g is a one-to-one function, then their pdfs have the following relationship:

$$f_Y(y) = \frac{f_X(x)}{|\mathrm{d}y/\mathrm{d}x|} = f_X(g^{-1}(y))\left|\frac{\mathrm{d}}{\mathrm{d}y}g^{-1}(y)\right| \quad \textit{or} \quad f_Y(y)|\mathrm{d}y| = f_X(x)|\mathrm{d}x|$$

***Proof*.**

By the definition, we have:

$$f_X(x) = \frac{\mathrm{d}}{\mathrm{d}x}F_X(x) \qquad F_X(x) = \mathbb{P}(X \leqslant x) \qquad f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y}F_Y(y) \qquad F_Y(y) = \mathbb{P}(Y \leqslant y) = \mathbb{P}[g(X) \leqslant g(x)]$$

Since g is one-to-one, it is strictly monotonic.

When $g' = \mathrm{d}y/\mathrm{d}x > 0$,

$$F_Y(y) = \mathbb{P}[g(X) \leqslant g(x)] = \mathbb{P}(X \leqslant x) = F_X(x)$$

$$\implies f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y}F_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y}F_X(x) = \frac{\mathrm{d}x}{\mathrm{d}y}\frac{\mathrm{d}}{\mathrm{d}x}F_X(x) = \frac{f_X(x)}{\mathrm{d}y/\mathrm{d}x}$$

When $g' = \mathrm{d}y/\mathrm{d}x < 0$,

$$F_Y(y) = \mathbb{P}[g(X) \leqslant g(x)] = \mathbb{P}(X \geqslant x) = 1 - F_X(x)$$

$$\implies f_Y(y) = -\frac{\mathrm{d}}{\mathrm{d}y}F_X(x) = -\frac{f_X(x)}{\mathrm{d}y/\mathrm{d}x}$$

$\square$

Definition 1.3 (Uniform Distribution)**.**

A CRV X is said to have a **uniform distribution** if its pdf is given by

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{for} \quad a \leqslant x \leqslant b \\ 0 & \text{otherwise} \end{cases}$$

and denoted as $X \sim U(a,b)$.

Theorem 1.2.

For $X \sim U(a,b)$,

- $\mathbb{E}[X] = \dfrac{a+b}{2}$
- $\operatorname{Var}(X) = \dfrac{(b-a)^2}{12}$
- $M_X(t) = \begin{cases} \dfrac{\mathrm{e}^{bt} - \mathrm{e}^{at}}{(b-a)t} & \textit{for} \quad t \neq 0 \\ 1 & \textit{for} \quad t = 0 \end{cases}$

Remark. $M_X(0) = 1$ makes it the continuous extension to $t = 0$.

Definition 1.4 (Mathmetical Expectation)**.**

Let $f(x)$ be the pdf of the CRV X with space S. Suppose g is a function of X, then

$$\mathbb{E}[g(X)] := \int_S g(x) f(x) \, \mathrm{d}x = \int_{-\infty}^{\infty} g(x) f(x) \, \mathrm{d}x$$

is called the **mathmetical expectation** or **expected value** of g, if the integral exists.

Remark. $g(X)$ gives another RV.

Definition 1.5 (Special Mathmetical Expectations)**.**

- Mean: $\mathbb{E}[X] = \displaystyle\int_S x f(x) \, \mathrm{d}x$
- Variance: $\operatorname{Var}(X) := \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] = \displaystyle\int_S (x - \mathbb{E}[X])^2 f(x) \, \mathrm{d}x$
- Moments: $\mathbb{E}[X^r] = \displaystyle\int_S x^r f(x) \, \mathrm{d}x$
- Mgf: $M_X(t) := \mathbb{E}\big[\mathrm{e}^{tX}\big] = \displaystyle\int_S \mathrm{e}^{tx} f(x) \, \mathrm{d}x \qquad -h < t < h$ for some $h > 0$

Definition 1.6 (The (100p)th Percentile)**.**

Given $p \in (0,1)$, π_p is a number such that

$$\mathbb{P}(X \leqslant \pi_p) = F_X(\pi_p) = \int_{-\infty}^{\pi_p} f_X(x) \, \mathrm{d}x = p$$

- The **first quartile** (the 25th percentile): $\quad q_1 := \pi_{0.25}$
- The **third quartile** (the 75th percentile): $\quad q_3 := \pi_{0.75}$
- The **median** or the **second quartile**: $\quad m := q_2 := \pi_{0.50}$

Definition 1.7 (The Upper 100α Percent Point)**.**

Given $\alpha \in (0,1)$, z_α is a number such that

$$\mathbb{P}(Z \geqslant z_\alpha) = 1 - F_Z(z_\alpha) = \alpha$$

Theorem 1.3.

z_α is the $100(1-\alpha)$th percentile of Z.

$$\mathbb{P}(Z \leqslant z_\alpha) = 1 - \alpha$$

2 The Exponential Distribution

Desciption Consider an APP on the time interval $[0, +\infty)$, with the average number of occurrences per unit time λ. Let the CRV X be the **waiting time** until the **first** occurrence.

Definition 2.1 (Exponential Distribution)**.**

Parameter $\theta > 0$

A CRV X has an **exponential distribution** with parameter $\theta > 0$ if its pdf is

$$f(x) = \frac{1}{\theta}\mathrm{e}^{-\frac{x}{\theta}} \quad \text{for} \quad x \geqslant 0, \theta > 0$$

where $\lambda = \dfrac{1}{\theta}$ is the average number of occurrences per unit time.

Theorem 2.1.

The cdf of X:

$$F(x) = 1 - \mathrm{e}^{-\frac{x}{\theta}} \quad \textit{for} \quad x \geqslant 0$$

Theorem 2.2.

- $M_X(t) = \mathbb{E}\left[\mathrm{e}^{tX}\right] = \dfrac{1}{1-t\theta} \quad \textit{for} \quad t < \dfrac{1}{\theta}$
- $\mathbb{E}[X] = \theta$
- $\mathrm{Var}(X) = \theta^2$

2.1 Properties

Theorem 2.3 (Memoryless Property)**.**

For $X \sim \mathrm{Exp}(\lambda)$,

$$\mathbb{P}(X > s + t | X > s) = \mathbb{P}(X > t) \qquad \forall s, t \geqslant 0$$

Definition 2.2 (Survival Function)**.**

The **survival function** of a non-negative CRV X is defined as

$$S(x) := \mathbb{P}(X > x) = 1 - F(x)$$

where $F(x)$ is the cdf of X.

Theorem 2.4.

For two independent exponential CRVs $X_1 \sim \operatorname{Exp}(\lambda_1)$ *and* $X_2 \sim \operatorname{Exp}(\lambda_2)$*, the minimum of* X_1 *and* X_2 *is also an exponential CRV:*

$$\min\{X_1, X_2\} \sim \operatorname{Exp}(\lambda_1 + \lambda_2)$$
$$\mathbb{P}(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2} \qquad \mathbb{P}(X_1 > X_2) = \frac{\lambda_2}{\lambda_1 + \lambda_2}$$

Proof.

For any independent CRVs X and Y,

$$\begin{aligned}
\mathbb{P}(X \leqslant Y) &= \int_{S_Y} \mathbb{P}(X \leqslant Y \mid Y = y)\mathbb{P}(Y = y) \\
&= \int_{S_Y} \mathbb{P}(X \leqslant y) f_Y(y)\,\mathrm{d}y \\
&= \int_{S_Y} F_X(y) f_Y(y)\,\mathrm{d}y
\end{aligned}$$

Plugging in the pdf and cdf of exponential CRVs, we have the answer. □

3 The Gamma Distribution

Description Consider an APP on the time interval $[0, +\infty)$, with the average number of occurrences per unit time λ. Let the CRV X be the **waiting time** until the $\boldsymbol{\alpha}$**-th** occurrence. The *exponential distribution* is a special case of the *gamma distribution*.

Proof.

Consider the cdf $F(x)$ when $x \geqslant 0$:

$$\begin{aligned}
1 - F(x) &= \mathbb{P}(X > x) \\
&= \mathbb{P}(\text{less than } \alpha \text{ occurrences over the time } [0, x]) \\
&= \sum_{n=0}^{\alpha-1} \frac{(\lambda x)^n \mathrm{e}^{-\lambda x}}{n!}
\end{aligned}$$

$$f(x) = F'(x) = \lambda \mathrm{e}^{-\lambda x} - \mathrm{e}^{-\lambda x} \sum_{n=1}^{\alpha-1} \left(\frac{\lambda(\lambda x)^{n-1}}{(n-1)!} - \frac{\lambda(\lambda x)^n}{n!} \right) = \frac{\lambda^\alpha x^{\alpha-1}}{(\alpha-1)!}\mathrm{e}^{-\lambda x} \quad \text{for} \quad x \geqslant 0$$

□

Definition 3.1 (The Gamma Function)**.**

(Generalized factorial)

$$\Gamma(x) := \int_0^\infty t^{x-1}\mathrm{e}^{-t}\,\mathrm{d}t \quad \text{for} \quad x > 0$$

$$\Gamma(x+1) = x\Gamma(x) \quad \text{for} \quad x > 0$$
$$(n-1)! = \Gamma(n) \quad \text{for} \quad n \in \mathbb{N}_+$$
$$\Gamma(\frac{1}{2} + n) = \frac{(2n-1)!!}{2^n}\sqrt{\pi}$$

Definition 3.2 (The Gamma Distribution)**.**

Parameters $\alpha > 0 \quad \theta > 0$

A CRV X has a **gamma distribution** with parameters $\alpha > 0, \theta > 0$ if its pdf is

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} \mathrm{e}^{-\frac{x}{\theta}} \quad \text{for} \quad x \geqslant 0, \alpha > 0, \theta > 0$$

where θ is the average waiting time for the first occurrences.

Remark. $\lambda = \dfrac{1}{\theta}$ is the average number of occurrences per unit time:

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \mathrm{e}^{-\lambda x} \quad \text{for} \quad x \geqslant 0, \alpha > 0, \lambda > 0$$

Theorem 3.1.

- $M_X(t) = \dfrac{1}{(1-\theta t)^\alpha} \quad \textit{for} \quad t < \dfrac{1}{\theta}$
- $\mathbb{E}[X] = \alpha\theta$
- $\mathrm{Var}(X) = \alpha\theta^2$

4 The Chi-Square Distribution

Description $\quad X \sim \chi^2(r) = \mathrm{Gamma}(\frac{r}{2}, \theta = 2) \quad \text{for} \quad r = 1, 2, 3, \cdots.$

Definition 4.1 (The Chi-Square Distribution)**.**

Parameter $r = 1, 2, 3, \cdots$

X has a **chi-square distribution** with degrees of freedom r if its pdf is

$$f(x) = \frac{1}{\Gamma\left(\frac{r}{2}\right) 2^{\frac{r}{2}}} x^{\frac{r}{2}-1} \mathrm{e}^{-\frac{x}{2}} \quad \text{for} \quad x \geqslant 0, r = 1, 2, 3, \cdots$$

and denoted as $X \sim \chi^2(r)$.

Theorem 4.1.

- $M_X(t) = (1-2t)^{-\frac{r}{2}} \quad \textit{for} \quad t < \dfrac{1}{2}$
- $\mathbb{E}[X] = r$
- $\mathrm{Var}(X) = 2r$

5 The Normal Distribution

Definition 5.1 (The Normal Distribution)**.**

Parameters $\mu \in \mathbb{R} \quad \sigma^2 \geqslant 0$

A CRV X is said to be normal or Gaussian if it has a pdf of the form

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad \text{for} \quad x, \mu \in \mathbb{R}, \quad \sigma \geqslant 0$$

and denoted as $X \sim N(\mu, \sigma^2)$.

Theorem 5.1.

- $M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right) \quad \textit{for} \quad t \in \mathbb{R}$
- $\mathbb{E}[X] = \mu$
- $\mathrm{Var}(X) = \sigma^2$

Definition 5.2 (The Standard Normal Distribution)**.**

$N(0,1)$ is called the **standard normal distribution**. If $Z \sim N(0,1)$, then the cdf of Z is denoted as $\Phi(z)$.

Theorem 5.2.

By the symmetry,

$$\Phi(-y) = 1 - \Phi(y) \quad \textit{for} \quad y \in \mathbb{R}$$

Theorem 5.3.

If $X \sim N(\mu, \sigma^2)$ *then*

$$Z = \frac{X-\mu}{\sigma} \qquad \Longrightarrow \qquad Z \sim N(0,1), \qquad Z^2 \sim \chi^2(1)$$
$$\mathbb{P}(a < X < b) = \mathbb{P}\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

Theorem 5.4.

If $Z \sim N(0,1)$ *then*

$$\mathbb{E}\left[Z^{2n-1}\right] = 0 \qquad \mathbb{E}\left[Z^{2n}\right] = (2n-1)!! \qquad \textit{for} \quad n = 1, 2, 3, \cdots$$

***Proof*.**

$$f_Z(z) = f_Z(-z) \quad \Longrightarrow \quad \mathbb{E}\left[Z^{2n-1}\right] = 0$$
$$Z^2 \sim \chi^2(1) \quad \Longrightarrow \quad \mathbb{E}\left[Z^{2n}\right] = \frac{\mathrm{d}^n}{\mathrm{d}t^n} M_{Z^2}(t)\bigg|_{t=0} = \frac{\mathrm{d}^n}{\mathrm{d}t^n}(1-2t)^{-\frac{1}{2}}\bigg|_{t=0} = (2n-1)!!$$

□

Theorem 5.5.

Let $X_1 \sim N(\mu_1, \sigma_2^2)$ *and* $X_2 \sim N(\mu_2, \sigma_2^2)$ *be two independent normal random variables. Then*

$$X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2) \qquad X_1 - X_2 \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$$

Let a *and* b *be two constants. Then*

$$aX_1 + bX_2 \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$$

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-1/2} \mathrm{e}^{-t}\,\mathrm{d}t = \int_0^\infty \frac{1}{u}\mathrm{e}^{-u^2}\,\mathrm{d}(u^2) = 2\int_0^\infty \mathrm{e}^{-u^2}\,\mathrm{d}u = \sqrt{\pi}$$
$$\Gamma(x+1) = x \cdot \Gamma(x) \quad \text{for} \quad x > 0 \qquad \Gamma(n + \frac{1}{2}) = \frac{(2n-1)!!}{2^n}\sqrt{\pi} \quad \text{for} \quad n = 1, 2, 3, \cdots$$

$$\int_{-\infty}^{\infty} \mathrm{e}^{-x^2}\,\mathrm{d}x = \sqrt{\pi} \qquad \int_{-\infty}^{\infty} \mathrm{e}^{-a(x+b)^2}\,\mathrm{d}x = \sqrt{\frac{\pi}{a}} \quad \text{for} \quad a > 0$$

$$\int_{-\infty}^{\infty} \mathrm{e}^{-a(\frac{x}{c}+b)^2}\,\mathrm{d}x = |c|\sqrt{\frac{\pi}{a}} \quad \text{for} \quad a > 0$$